

Exploratory Data Analysis (EDA) Summary Report

1.Introduction:

This report presents an exploratory data analysis (EDA) of the Delinquency Prediction Dataset. The objective is to assess data quality, identify missing values and anomalies, and uncover key factors that may influence customer delinquency risk.

2.Data Set Overview:

Number of Records: 1,000

Key Variables: Customer_ID, Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Delinquent_Account, Loan_Balance, Debt_to_Income_Ratio, Employment_Status, Account_Tenure, Credit_Card_Type, Location, Month_1–Month_6 Payment History.

Data Types:

Numerical: Age, Income, Credit Score, Credit Utilization, Missed Payments, Loan Balance, Debt-to-Income Ratio, Account Tenure

Categorical: Employment Status, Credit Card Type, Location, Payment History

Binary: Delinquent Account

Anomalies Observed: Potential outliers were identified in Income, Loan Balance, Credit Utilization, and Debt-to-Income Ratio fields.

3. Missing Data Analysis:

Variables with Missing Values:

Income (39 missing values)

Loan_Balance (29 missing values)

Credit_Score (2 missing values)

Missing Data Treatment:

Median imputation for Income and Loan_Balance.

Mean/Median imputation for Credit_Score.

Missing values were retained and imputed rather than deleting records to preserve data.

4. Key Findings and Risk Indicators:

Correlations Observed Between Key Variables:

Higher Missed_Payments are associated with greater delinquency risk.

Lower Credit_Score is linked to increased probability of delinquency.

Higher Debt_to_Income_Ratio and Credit_Utilization indicate financial stress and higher risk.

Frequent "Late" or "Missed" payment statuses are strong indicators of future delinquency.

Unexpected Anomalies:

Extremely high income and loan balance values.

Customers with unusually high credit utilization rates.

Repeated missed payments across multiple months.

Top Risk Indicators:

Missed_Payments

Credit_Score

Debt_to_Income_Ratio

Credit_Utilization

Employment_Status

5.AI & GenAI Usage:

AI Prompts Used:

Summarize key patterns, outliers, and missing values in the dataset.

Suggest an imputation strategy for missing income values.

Identify the top variables most likely to predict delinquency.

Highlight risk factors affecting delinquency prediction.

AI-Assisted Insights:

Identification of missing data handling methods.

Detection of potential outliers and anomalies.

Prioritization of key delinquency risk indicators.

6.Conclusion & Next Steps:

The analysis identified missing values in Income, Loan_Balance, and Credit_Score, which were addressed through imputation techniques. Key predictors of delinquency include Missed_Payments, Credit_Score, Debt_to_Income_Ratio, and Credit_Utilization. Further analysis and predictive modeling can be conducted to improve delinquency forecasting and support risk management decisions.